

# IEEE-Style Tables and Graphs for MXFP4 GDN Decode

Table 1: End-to-End Decode Performance

| Platform          | Format    | Latency (us) | Power (W) | Speedup |
|-------------------|-----------|--------------|-----------|---------|
| NVIDIA H100 PCIe  | FP32/BF16 | 285.0        | 350.0     | 1.0x    |
| USC FPGA Baseline | BF16      | 63.2         | <150.0    | 4.5x    |
| Ours (Alveo U55C) | MXFP4     | 51.788       | 9.096     | 5.5x    |

Table 2: Post-Implementation Resource Utilization

| Resource | Utilization | Notes                 |
|----------|-------------|-----------------------|
| LUT      | 6.64%       | Native E2M1 multiply  |
| DSP      | 7.27%       | Fixed-point alignment |
| BRAM     | 3.77%       | Persistent state      |

Table 3: Parallelism and Block-Size Sweep

| Config (B, P_K, P_V) | Latency (us) | LUT    | DSP    | BRAM  | Power    |
|----------------------|--------------|--------|--------|-------|----------|
| (32,16,8) Default    | 51.788       | 6.64%  | 7.27%  | 3.77% | 9.096 W  |
| (16,32,16)           | 8240.176     | 24.96% | 28.72% | 0.57% | 11.004 W |
| (32,32,16)           | 8240.176     | 24.96% | 28.72% | 1.57% | 11.004 W |

Table 4: Synthetic Quantization Ablation

| Format     | Synthetic Output Cosine | State Relative L2 Error |
|------------|-------------------------|-------------------------|
| FP32       | 1.000                   | 0.000                   |
| Flat INT4  | 0.967                   | --                      |
| MXFP4 B=16 | 0.982292                | 0.117084                |
| MXFP4 B=32 | 0.981823                | 0.117550                |

Table 5: Per-Tensor Sensitivity Profile

| Tensor | Output Cosine Similarity |
|--------|--------------------------|
| Value  | 0.999580                 |
| Key    | 0.999523                 |
| Gate   | 0.995668                 |
| State  | 0.993845                 |
| Query  | 0.992969                 |

Figure 1: Pareto View

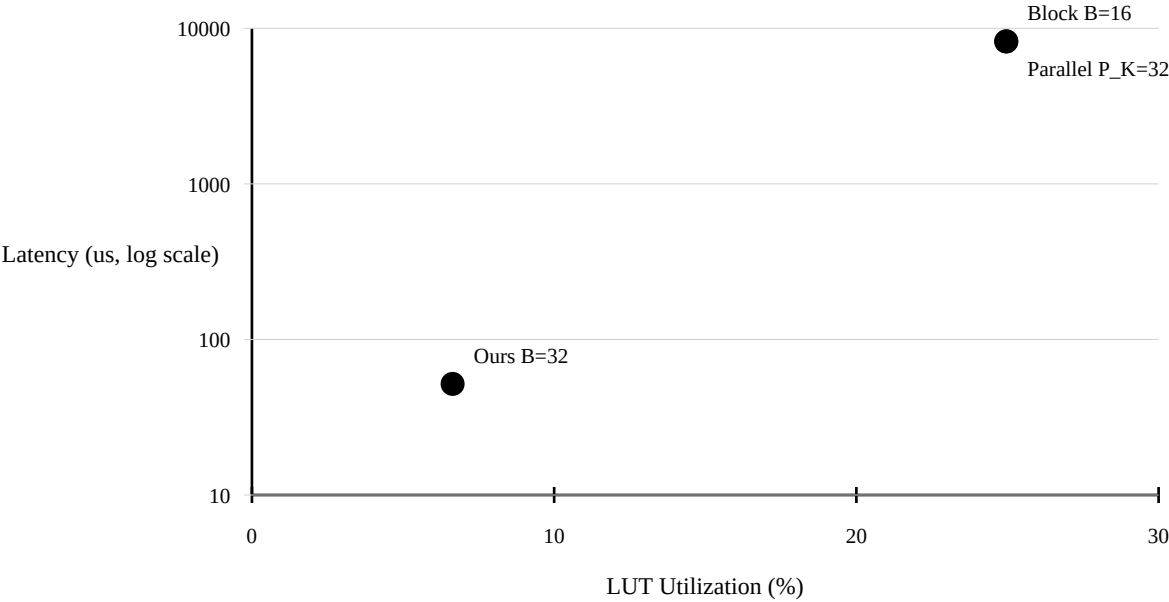


Figure 2: Accuracy

